



ABHISHEK KUMAR

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PROFESSIONAL SUMMARY

M.Tech Computer Science student at IIT Tirupati with a strong foundation in Machine Learning, Generative AI, and Data Engineering. Experienced in developing LLM-powered applications, AI automation systems, and financial time-series models using Python, PyTorch, FastAPI, and SQL. Hands-on experience in building scalable AI solutions, multi-agent workflows, and data-driven applications with a passion for solving real-world business problems through artificial intelligence.

EDUCATION

Degree/Certificate	Institute/Board	Year
M.Tech. (CSE)	IIT Tirupati	2026
B.Tech (CSE)	IIT ISM Dhanbad	2020

EXPERIENCE

- AdzyNow Pvt Ltd** Aug. 2025 – Present
Data Engineer Intern
Dhanbad, India
 - Developed Python-based ETL pipelines to ingest, preprocess, and transform Meta Ads and Google Ads campaign data into automated analytics dashboards, reducing report generation time by 70%.
 - Automated campaign reporting workflows, saving 8–10 hours/week through data pipeline optimization.
 - Analyzed 20+ marketing KPIs to generate data-driven optimization insights for client campaigns.

PROJECTS

- Autonomous Research Agent** Jan 2026 - Present
Developed an AI agent that auto-generates and runs scientific experiments with LLMs.
 - Designed and developed a multi-agent research pipeline where LLMs automatically create hypotheses, write experiment code, run it in isolated virtual environment, and iteratively refine results based on execution feedback.
 - Implemented automated dependency resolution and experiment sandboxing, enabling each experiment to dynamically detect required Python libraries, install them, and execute safely without human intervention.
 - Integrated a unified LLM client to ensure all experiments consistently use the project's core model, supporting advanced tasks such as RAG, evaluation, and code generation.
 - Tools & Technologies:** Python, LLMs(OpenAI/Gemini), Multi-Agent Systems, Automation, Virtual Environments, Dependency Management.
- Generative Modeling of Financial Time Series using TimeGAN and Diffusion Models** Aug. 2025 - Present [Github](#)
TimeGAN and (DDPM) to generate synthetic financial time series preserving temporal dependencies.
 - Constructed end-to-end machine learning pipelines on 20+ years of S&P 500 and GOOG time-series data, including preprocessing, normalization, and sliding-window feature engineering.
 - Trained TimeGAN and Diffusion models to generate synthetic financial sequences preserving temporal correlations and distributional statistics.
 - Validated synthetic data realism using DTW distance and Train-on-Synthetic-Test-on-Real (TSTR), achieving comparable predictive performance to real data.
 - Tools & Technologies:** Python, PyTorch, TimeGAN, Diffusion Models (DDPM), LSTM, NumPy, Pandas, DTW, TSTR.
- Finsphere (App)** Jan. 2025 - May. 2025 [Github](#)
Fintech solution to address financial inclusion by integrating mobile banking, microfinance.
 - Designed and delivered a cross-platform fintech application integrating mobile banking, microfinance, and digital insurance modules.
 - Integrated REST APIs and structured data schemas to support secure transactions, validation, and fault-tolerant workflows.
 - Improved reliability of financial operations by introducing centralized error handling and schema-based validation.
 - Tools & Technologies:** React Native, Hooks, Axios, REST APIs, Gesture Handler, Reanimated

TECHNICAL SKILLS

- Programming:** C/C++, Python, JavaScript, SQL
- AI/ML:** NLP, LLMs, RAG, Prompt Engineering, Multi-Agent Systems, TimeGAN, Diffusion Models
- Tools & OS:** Git, GitHub, Linux, Windows, Mac
- Web Skills/Frameworks:** HTML5, CSS, Tailwind CSS, React, FastAPI, REST APIs
- Library & Frameworks:** Pandas, Numpy, Matplotlib, Seaborn, PyTorch
- Core Concepts:** Data Structure, Algorithms, OOPs, DBMS, Computer Networks
- Cloud:** AWS